AWS Static Website Hosting using Terraform

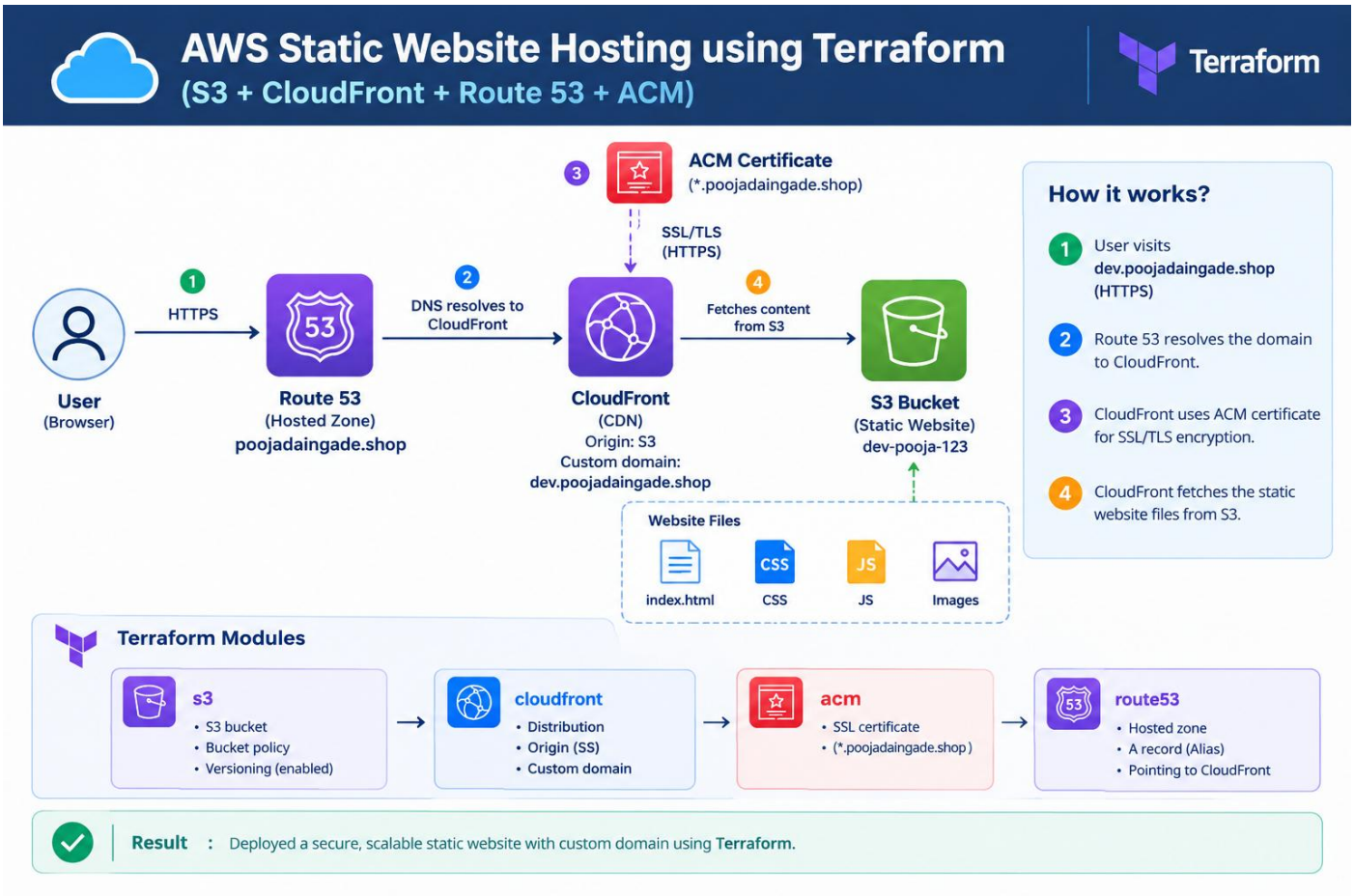
Prepared By: Pooja Daingade

Project Overview

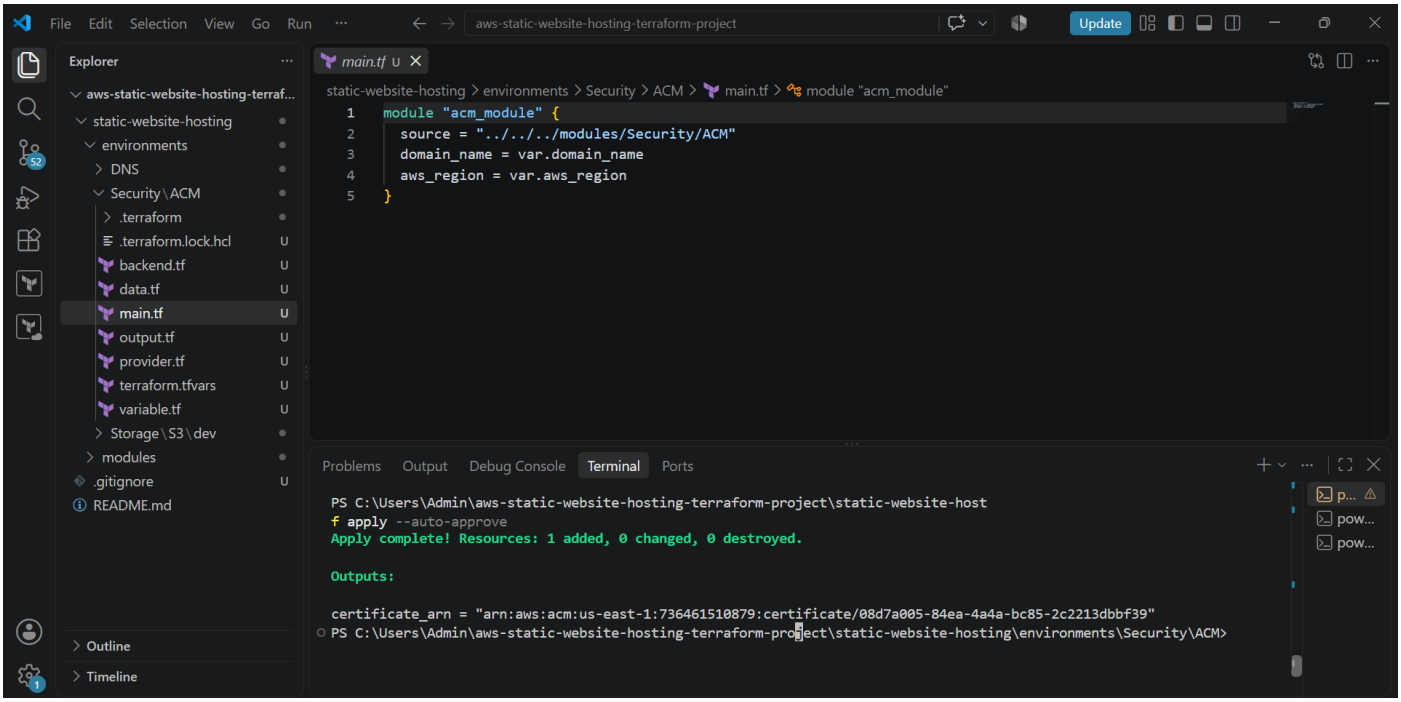
This project demonstrates how to host a static website on AWS using Terraform. It uses Amazon S3 to store website files, Amazon CloudFront for content delivery, Amazon Route 53 for custom domain configuration, and AWS Certificate Manager (ACM) to enable HTTPS.

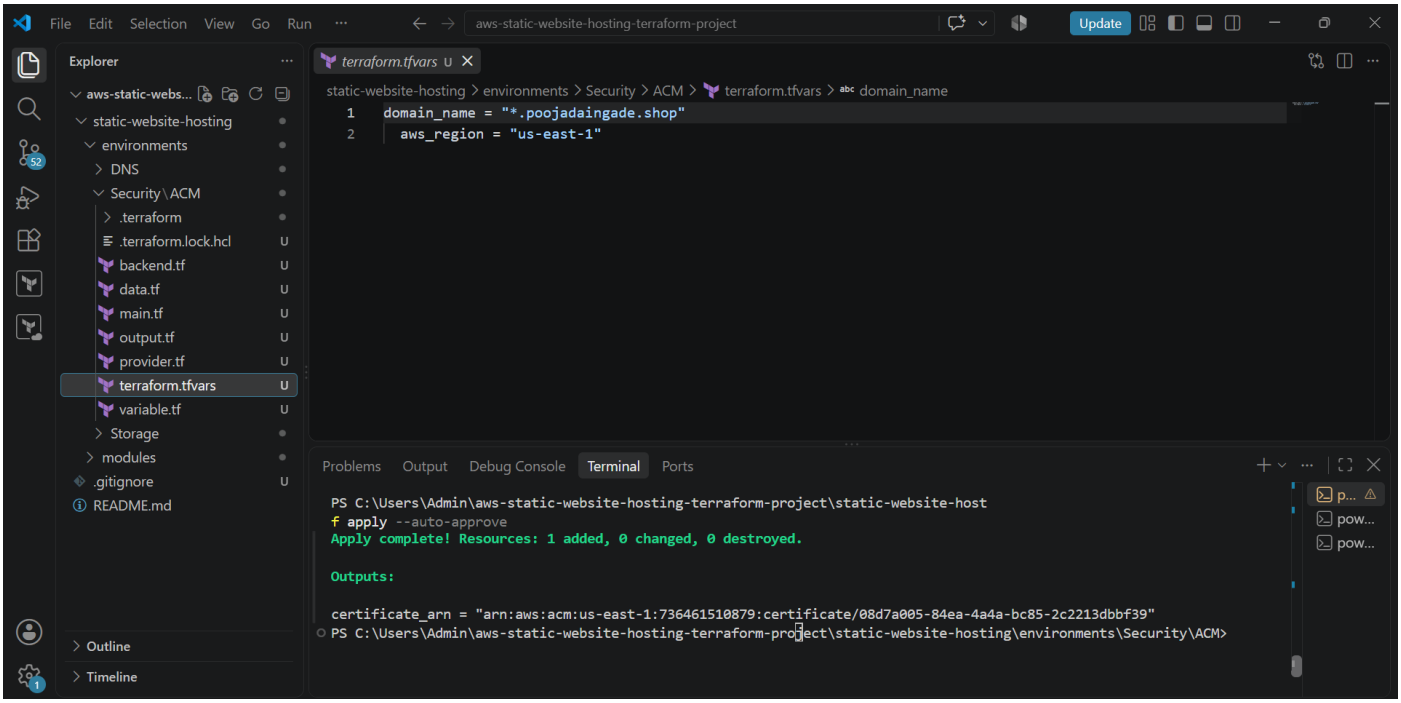
Terraform modules are used to provision and manage the AWS infrastructure in a structured and reusable way. The website is accessible through the custom domain dev.poojadaingade.shop with secure HTTPS access.

Architecture Diagram

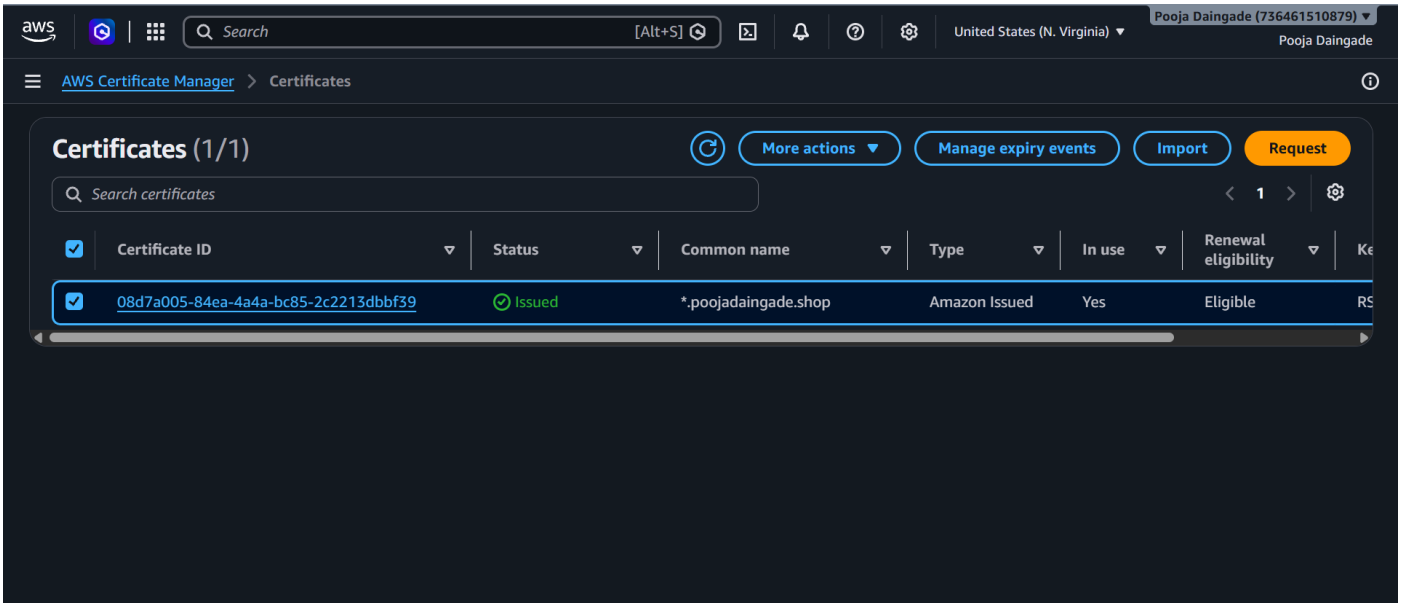


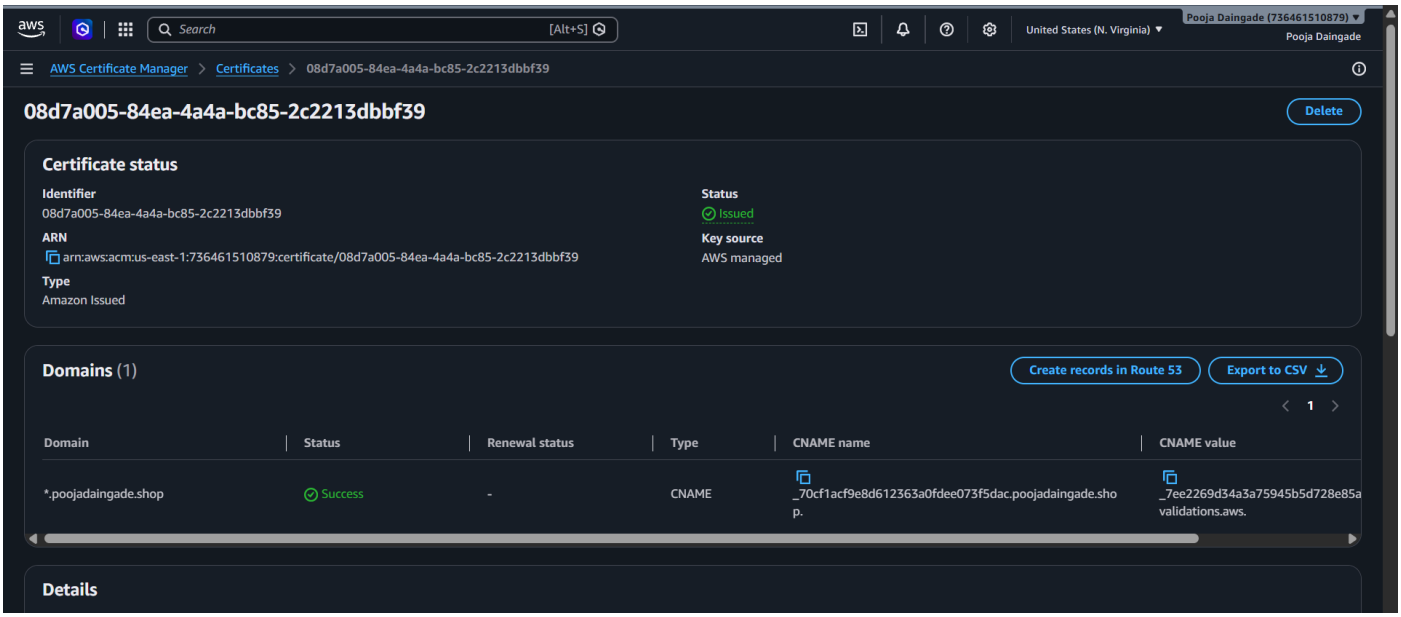
1. Successfully created the ACM certificate using Terraform and retrieved the certificate ARN through outputs. Configured the domain ***.poojadaingade.shop** and AWS Region **us-east-1** using Terraform variables.



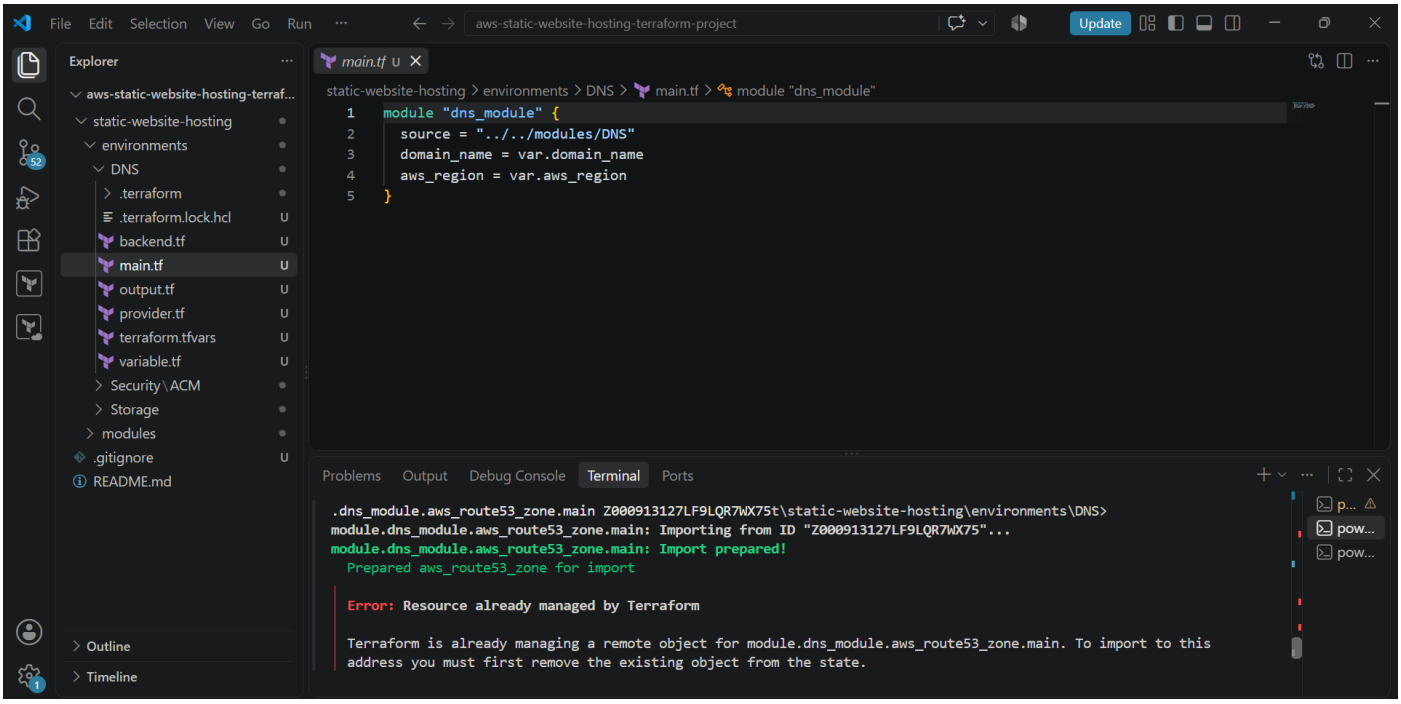


2. Verified the certificate status as **Issued** and confirmed successful validation for *.poojadaingade.shop in us-east-1.

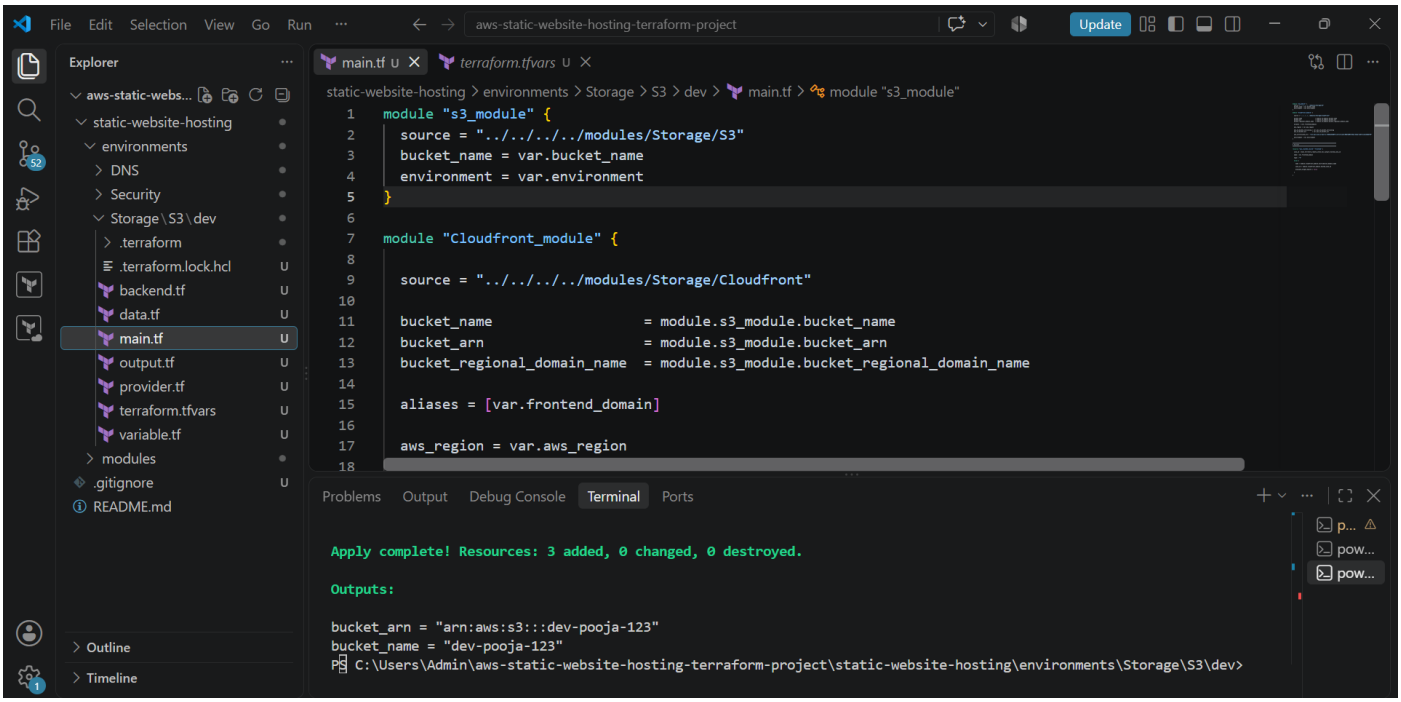


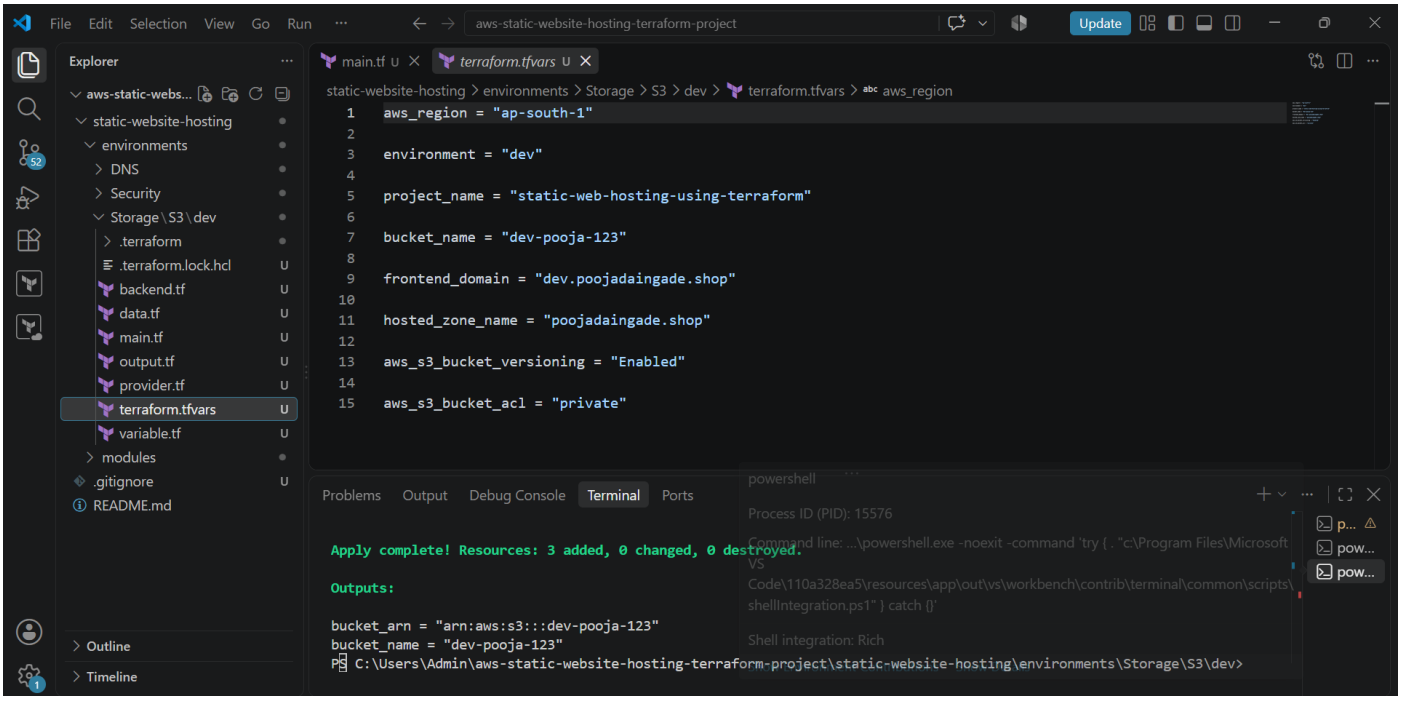


3. Imported the existing AWS Route 53 Hosted Zone into Terraform using the import command and configured it through a reusable DNS module.
   tf import module.dns_module.aws_route53_zone.main Z000913127LF9LQR7WX75

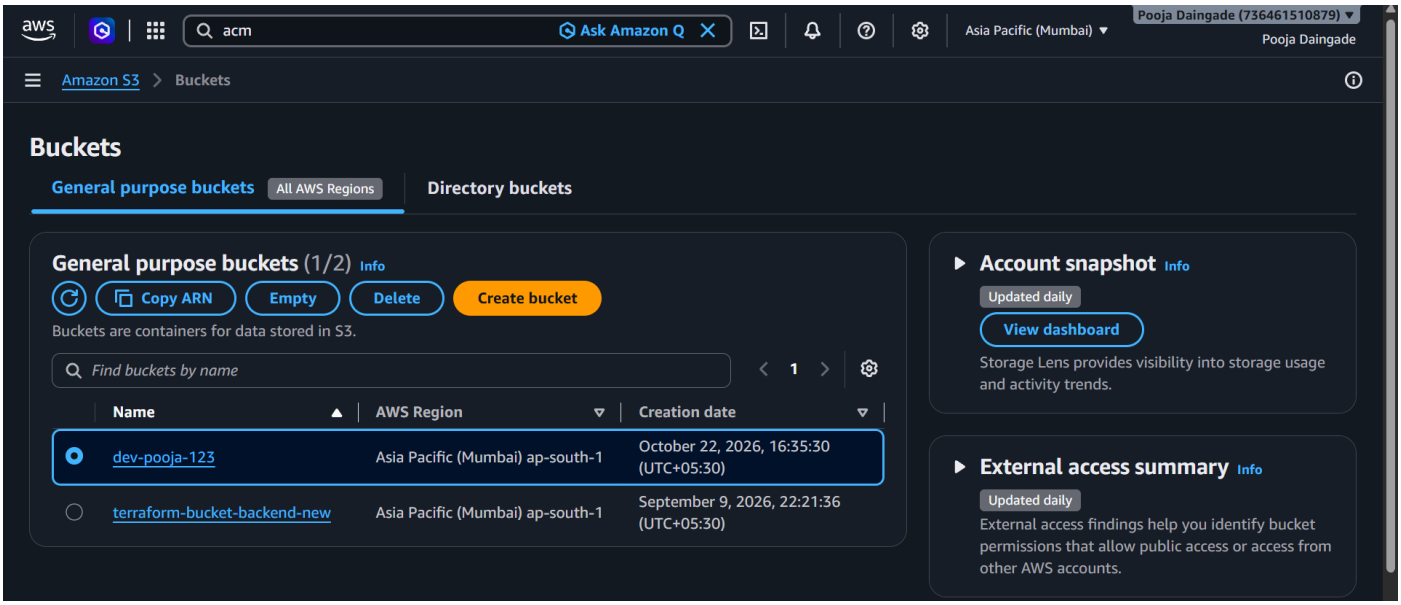


4. Configured an S3 bucket in **ap-south-1** with versioning enabled and private access. Integrated the CloudFront distribution with the custom domain **dev.poojadaingade.shop** and ACM SSL certificate using reusable Terraform modules. Successfully provisioned the resources.

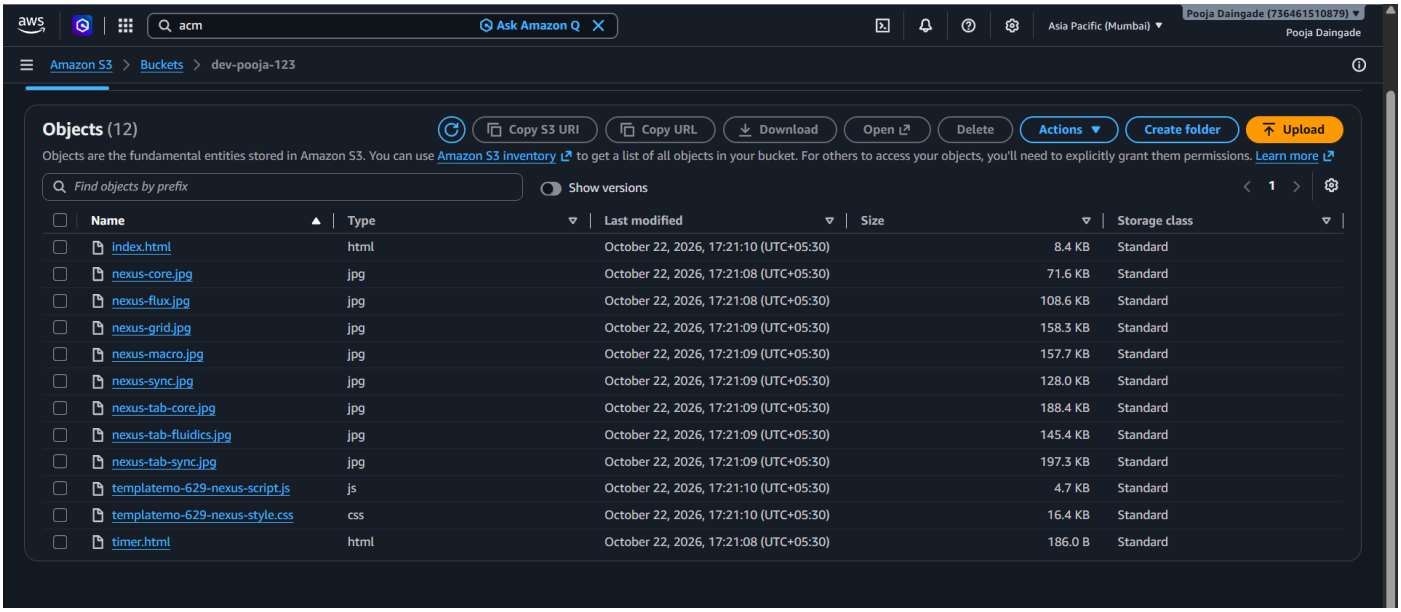




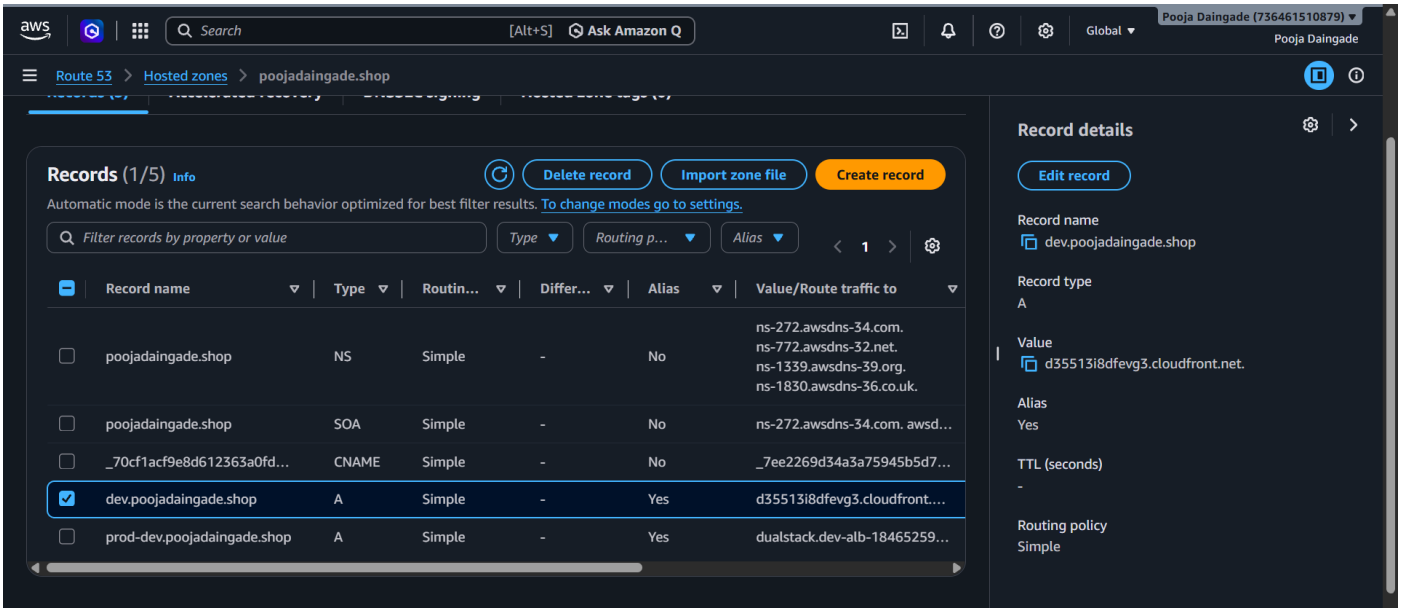
5. Verified the S3 bucket **dev-pooja-123** in the AWS Management Console. The bucket was successfully created in the Mumbai region (ap-south-1) using Terraform.



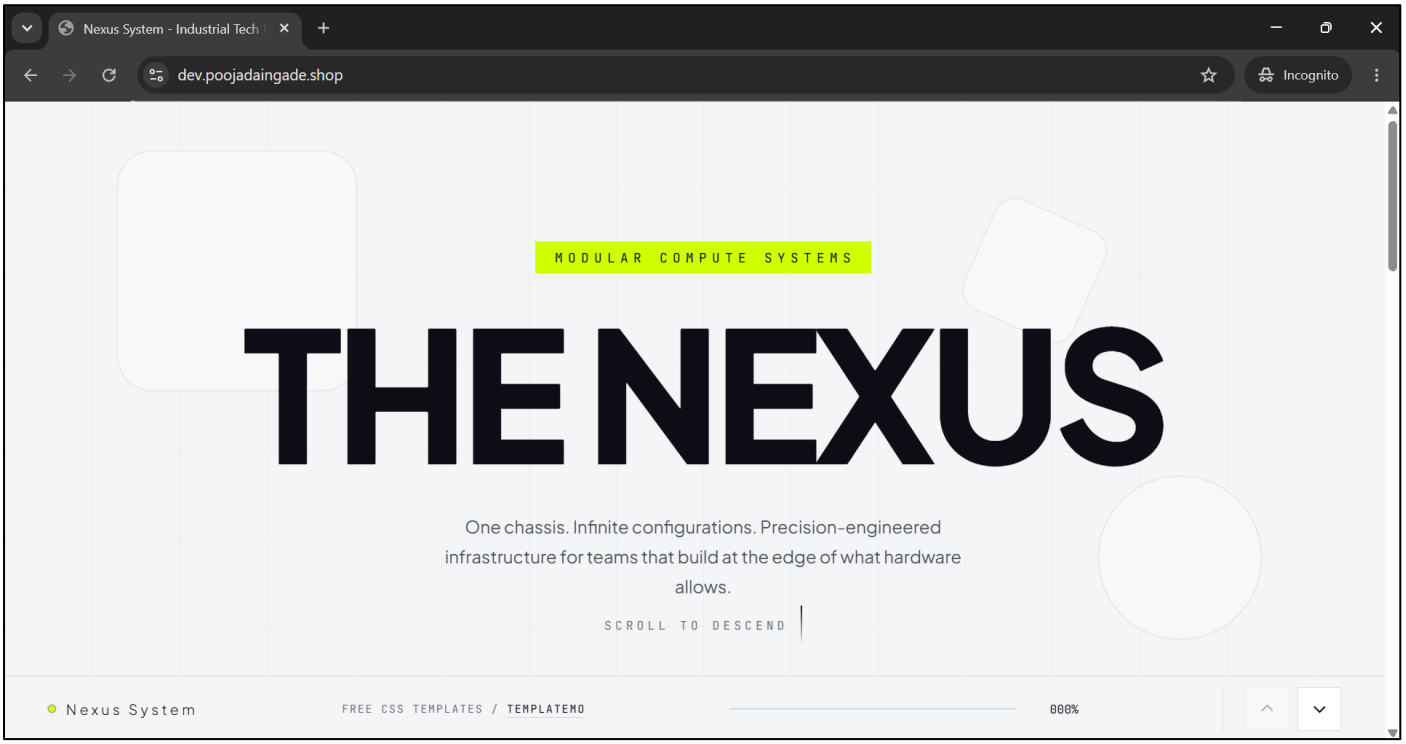
6. The bucket contains the static website files, including index.html, CSS, JavaScript, and images, required for website hosting.



7. Verified the Route 53 hosted zone poojadaingade.shop in the AWS Console. The A record for dev.poojadaingade.shop is configured as an Alias pointing to the CloudFront distribution.



8. Successfully verified the static website at **dev.poojadaingade.shop** in the browser. The website is accessible over HTTPS and displays the Nexus System homepage, confirming successful website delivery through CloudFront.



Conclusion

This project provided hands-on experience in hosting a static website on AWS using Terraform. It helped me understand Infrastructure as Code, Terraform modules, AWS services, custom domain configuration, and secure HTTPS access.